

How Directed Acyclic Graphs (DAGs) Help Represent Causal Relationships

An Overview

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Introduction

- ▶ **Correlation is not causation**, yet many important research questions inevitably involve drawing causal conclusions.
- ▶ In many fields (e.g., psychology, social sciences), **true randomization is not always feasible or ethical** (e.g., we cannot randomly assign individuals to different socioeconomic statuses).
- ▶ Researchers often must rely on **observational data**, where hidden confounders, measurement errors, or selection biases can lead to misleading inferences.
- ▶ **Directed Acyclic Graphs (DAGs)** offer a **clear, visual, and flexible** way to represent causal assumptions. By mapping out who influences whom, they help us:
 - ▶ Identify potential confounders (common causes of two variables),
 - ▶ Distinguish mediators (intervening pathways) from colliders (common effects),
 - ▶ Avoid overcontrolling for variables that might introduce new biases.
- ▶ Ultimately, DAGs provide a transparent framework to **guide empirical analysis**—from deciding which variables to measure or control, to interpreting results more cautiously.

Directed Acyclic Graphs Principles

- ▶ **Directed:** Arrows indicate a causal direction ($A \rightarrow B$ means “A causes B”).
- ▶ **Acyclic:** No feedback loops; a variable cannot cause itself, even indirectly.
- ▶ **Nodes** represent variables (e.g., Intelligence, Income).
- ▶ **Edges (Arrows)** capture hypothesized causal relationships among variables.

Key Benefit: DAGs allow us to visualize how variables interrelate, helping to:

- ▶ Identify *confounders* (common causes),
- ▶ Distinguish *mediators* (on the causal pathway),
- ▶ Recognize *colliders* (common effects that should not be controlled).

Common DAG Building Blocks

Chain: $X_1 \rightarrow X_2 \rightarrow X_3$



Fork: $X_1 \leftarrow X_2 \rightarrow X_3$



Collider: $X_1 \rightarrow X_2 \leftarrow X_3$



(a) Chain:
 $X_1 \rightarrow X_2 \rightarrow X_3$

(b) Fork:
 $X_1 \leftarrow X_2 \rightarrow X_3$

(c) Collider:
 $X_1 \rightarrow X_2 \leftarrow X_3$

- ▶ **Chain** can transmit a causal effect from X_1 to X_3 via X_2 .
- ▶ **Fork** indicates a common cause (X_2) that can confound the relationship between X_1 and X_3 .
- ▶ **Collider** normally blocks correlation between X_1 and X_3 unless you condition on X_2 , which can induce “collider bias.”

Figure 1: A Simple DAG

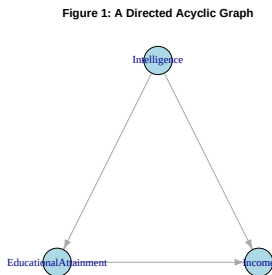


Figure 1: A basic three-variable DAG

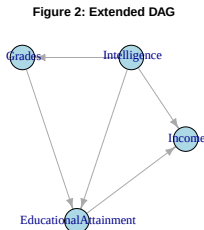
Nodes and Arrows:

- ▶ **Intelligence** → **Educational Attainment**
- ▶ **Intelligence** → **Income**
- ▶ **Educational Attainment** → **Income**

Interpretation:

- ▶ Intelligence influences both educational attainment and income.
- ▶ Educational attainment affects income directly.
- ▶ Intelligence could be a confounder if we only focus on education → income.

Figure 2: An Extended DAG



(Figure 2: Intelligence, Grades, Educational Attainment, and Income.)

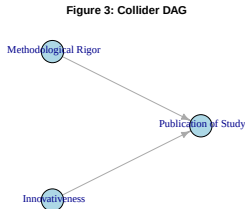
Added Node: Grades

- ▶ **Intelligence** → **Grades**
- ▶ **Grades** → **Educational Attainment**
- ▶ Intelligence can still directly affect both Educational Attainment and Income.

Key Points:

- ▶ Multiple causal pathways: Intelligence may reach Income via Grades & Education.
- ▶ Mediator vs. Confounder: Grades likely acts as a mediator, not a confounder.
- ▶ Chain/Fork/Collider: Longer paths in this DAG can be broken down into $(A \rightarrow B \rightarrow C)$, $(A \leftarrow B \rightarrow C)$, or $(A \rightarrow B \leftarrow C)$.
- ▶ Controlling Variables: Deciding which variables to control depends on their role (confounder, mediator, or collider).

Figure 3: Collider DAG



- ▶ A collider occurs when two arrows head into the same node
- ▶ Avoid conditioning on the collider to prevent bias

Key Advantages

1. **Visualizing Causal Structure:**

- ▶ Clarifies which nodes are potential confounders, mediators, or colliders.

2. **Guiding Control Strategies:**

- ▶ Identifies which variables to adjust for in regression, matching, or stratification.

3. **Avoiding Bias:**

- ▶ Prevents accidental control of colliders or mediators, which could distort results.

4. **Flexible Function Forms:**

- ▶ DAGs do not assume linearity; an arrow simply indicates “cause $\rightarrow$ effect.”

Conclusion

- ▶ **DAGs** are a powerful **qualitative tool** for representing causal assumptions. In summary, in observational studies, DAGs can clearly visualize causal hypotheses, help us identify and distinguish key causal structures such as confounding, collider, and mediator, and provide ideas and direction for more accurate and robust causal inference.
- ▶ They help us see *who causes what*, and highlight *how* to handle different variables (confounders vs. mediators vs. colliders).
- ▶ Always combine DAG reasoning with domain knowledge and sound methods (e.g., proper measurement, robust design).

Thank you!

Questions?